

Notation used throughout

Ω	sample space
A, B	events
A^c	complement of A
X, Y	random variables
x_i, y_i	observed values in row i
n_A, n, N	count in A , sample size, population size
μ, σ^2, p	population mean, variance, proportion
$\bar{X}, S^2, \hat{P}$	sample mean, variance, proportion estimators
$\bar{x}, s^2, \hat{p}$	corresponding realised estimates
θ, θ_0	parameter of interest, value specified by H_0
b_j, β_j	estimated coefficient, population coefficient
$SE(\cdot)$	standard error
α	significance level
df	degrees of freedom

Two-sided hypothesis test: $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Lecture 1 - Data summaries and units

Proportions and percentages

$$\text{proportion} = \frac{n_A}{n}, \quad \text{percentage} = 100 \frac{n_A}{n} \%.$$

Sample mean and range

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \text{range} = \max_i x_i - \min_i x_i.$$

Changes between two percentages a and b

$$\text{percentage-point change} = b - a,$$

$$\text{relative percentage change} = 100 \frac{b-a}{a} \% \quad (a \neq 0).$$

Lecture 2 - Probability foundations

Probability bounds, sample space, and complement

$$0 \leq P(A) \leq 1, \quad P(\Omega) = 1, \quad P(A^c) = 1 - P(A).$$

Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Addition rule for disjoint events

$$A \cap B = \emptyset \implies P(A \cup B) = P(A) + P(B).$$

Equally likely outcomes

$$P(A) = \frac{|A|}{|\Omega|} \quad (\text{equally likely outcomes}).$$

Random variable

$$X : \Omega \rightarrow \mathbb{R}.$$

Partition of the sample space

$$B_i \cap B_j = \emptyset \quad (i \neq j), \quad \bigcup_{i=1}^k B_i = \Omega.$$

Lecture 3 - PMFs, CDFs, and expectation

Probability mass function (PMF)

$$p_X(x) = P(X = x), \quad p_X(x) \geq 0, \quad \sum_{x \in \mathcal{X}} p_X(x) = 1.$$

Cumulative distribution function (CDF)

$$F_X(a) = P(X \leq a) = \sum_{x \leq a} p_X(x).$$

Interval probability from a CDF

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Expectation and expectation of a transformation

$$E[X] = \sum_{x \in \mathcal{X}} x p_X(x), \quad E[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x).$$

Bernoulli distribution

$$X \sim \text{Bernoulli}(p) : \quad P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad E[X] = p.$$

Lecture 4 - Variance and conditioning

Population variance and standard deviation

$$\mu = E[X], \quad \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \{E[X]\}^2.$$

$$\text{SD}(X) = \sqrt{\text{Var}(X)}.$$

Bernoulli variance

$$X \sim \text{Bernoulli}(p) : \quad \text{Var}(X) = p(1 - p).$$

Sample variance and standard deviation

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

Conditional probability and product rule

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A | B)P(B), \quad P(B) > 0.$$

Law of total probability

$$P(A) = \sum_{i=1}^k P(A | B_i)P(B_i).$$

Bayes' rule

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^k P(A | B_j)P(B_j)}.$$

Lecture 5 - Joint PMFs and correlation

Joint PMF

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

Marginal and conditional PMFs

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Independence of random variables

$$X \perp Y \iff p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y.$$

Independence of events

$$A \perp B \iff P(A \cap B) = P(A)P(B).$$

Population covariance and correlation

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y].$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}, \quad -1 \leq \rho_{XY} \leq 1.$$

Sample covariance and correlation

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

Lecture 6 - Continuous distributions and quantiles

Continuous-distribution probabilities

$$f_X(x) \geq 0, \quad P(X=x) = 0, \quad P(a < X \leq b) = F_X(b) - F_X(a).$$

Uniform distribution

$$X \sim U(a, b) : \quad f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

$$P(c < X < d) = \frac{d-c}{b-a}, \quad E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Quartiles and interquartile range

$$Q_1 = q_{0.25}, \quad Q_2 = q_{0.50}, \quad Q_3 = q_{0.75}, \quad \text{IQR} = Q_3 - Q_1.$$

Normal distribution and standardisation

$$X \sim N(\mu, \sigma^2), \quad Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68, \quad P(\mu - 1.96\sigma \leq X \leq \mu + 1.96\sigma) \approx 0.95.$$

Lecture 7 - Samples and estimators

Sample mean and sample proportion estimators

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{P} = \frac{1}{n} \sum_{i=1}^n X_i \quad (X_i \in \{0, 1\}).$$

Sample variance and degrees of freedom

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \sum_{i=1}^n (X_i - \bar{X}) = 0, \quad df = n - 1.$$

10% condition for sampling without replacement

$$n \leq 0.10N \quad (\text{sampling without replacement}).$$

Lecture 8 - Sampling distributions and SEs

Sampling distribution of the sample mean

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{S}{\sqrt{n}}.$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

One-sample t statistic

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Sampling distribution of a sample proportion

$$E[\hat{P}] = p, \quad \text{SE}(\hat{P}) = \sqrt{\frac{p(1-p)}{n}}.$$

Success-failure condition

$$np \geq 10, \quad n(1-p) \geq 10.$$

Lecture 9 - Confidence intervals and tests

General confidence-interval form

$$\text{estimate} \pm \text{critical value} \times \text{SE}.$$

One-sample t confidence interval for a mean

$$\bar{X} \pm t_{1-\alpha/2, n-1} \frac{S}{\sqrt{n}}.$$

Critical values for a 95% confidence interval

$$z_{0.975} = 1.960, \quad \begin{array}{c|ccccc} df & 5 & 10 & 20 & 30 & 60 & \infty \\ \hline t_{0.975, df} & 2.571 & 2.228 & 2.086 & 2.042 & 2.000 & 1.960 \end{array}$$

Difference from a hypothesised mean

$$\delta = \mu - \mu_0, \quad \hat{\delta} = \bar{X} - \mu_0, \quad T = \frac{\hat{\delta}}{S/\sqrt{n}}, \quad df = n - 1.$$

Two-sided hypothesis test for a mean

$$H_0 : \delta = 0 \quad \text{versus} \quad H_1 : \delta \neq 0.$$

Two-sided p -value and decision rule

$$p = P(|T_{df}| \geq |t_{obs}| \mid H_0), \quad p \leq \alpha \implies \text{reject } H_0.$$

Lecture 10 - Tests, errors, and power

Two-sided hypothesis test for a proportion

$$H_0 : p = p_0 \quad \text{versus} \quad H_1 : p \neq p_0.$$

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \quad p\text{-value} = 2P(Z \geq |z_{obs}|).$$

Success-failure condition under H_0

$$np_0 \geq 10, \quad n(1-p_0) \geq 10.$$

Confidence interval for a proportion

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Type I error and power

$$P(\text{Type I error}) = \alpha, \quad \text{Power} = P(\text{reject } H_0 \mid H_1 \text{ is true}).$$

Two-sided test and confidence interval for a regression coefficient

$$t = \frac{b_j}{\text{SE}(b_j)}, \quad b_j \pm t_{1-\alpha/2, df} \text{SE}(b_j).$$

$$H_0 : \beta_j = 0 \quad \text{versus} \quad H_1 : \beta_j \neq 0.$$